

ITDMA Assignment 1 Working Pack

Risk Mitigation in CI/CD Pipelines for Safety-Critical Embedded Systems

Referencing style: Harvard author-date | Format target: Arial 12, 1.5 spacing | Due: 8 June 2026, 23:59

How to Use This Pack

- Use this document as a working pack and writing guide, not as a final submission.
- Write the final article in your own words, using the evidence prompts and Harvard citation cues.
- Replace every bracketed writing prompt before submission.
- Keep a live references list while drafting so in-text citations and reference entries match.

Research Focus

Topic: Risk Mitigation in CI/CD Pipelines for Safety-Critical Embedded Systems Using Automated Testing and Rollback Strategies.

Research question: How can automated testing and rollback strategies within CI/CD pipelines reduce deployment risks in safety-critical embedded systems?

Objectives:

1. Identify key risks associated with automated deployments in safety-critical systems.
2. Examine automated testing techniques suitable for embedded CI/CD pipelines.
3. Design a risk-aware CI/CD pipeline incorporating rollback and fail-safe mechanisms.

Distinction Argument

The strongest argument is that safety-critical embedded CI/CD cannot simply copy web-style continuous deployment. It needs layered verification, traceable certification evidence, staged deployment, monitoring, and recovery mechanisms that select between rollback, fail-safe, fail-silent, fail-operational and graceful degradation modes.

Article Structure and Word Budget

Section	Words	Writing instruction
Abstract	150-200	Write last. State the topic, method, core finding and recommendation.
Introduction	350-450	Problem, objectives, research question and roadmap. Use Zampetti et al. (2022), Dakkak et al. (2023), Baron and Louis (2023).
Methodology	450-600	Databases, search strings, inclusion/exclusion criteria and matrix fields.
Literature Review	650-800	Theme-based synthesis of CI/CD, testing, certification and fail-safe recovery.
Results / Comparative Analysis	800-1000	Compare paper structures, methodologies, approaches, data collection, sampling, analysis and findings.
Discussion	600-800	Synthesis, gaps, implications and pipeline design response.
Conclusion and Venue	350-500	Answer research question, recommendations and ICST venue fit.

Common Research Paper Sections

Section	Purpose	Typical content	Style	Citation cue
Abstract	Summarises the topic, method, key	Problem context, literature-search	Concise, objective and written after the	Usually no citations unless your lecturer

Section	Purpose	Typical content	Style	Citation cue
	synthesis and contribution.	method, main findings, gaps and recommendation.	article is complete.	expects them; focus on your synthesis.
Introduction	Frames why CI/CD risk mitigation matters in safety-critical embedded systems.	Context, problem, objectives, research question and article roadmap.	Persuasive but evidence-based; move from broad problem to focused question.	Use sources such as Zampetti et al. (2022), Dakkak et al. (2023) and Baron and Louis (2023).
Literature Review	Synthesises existing knowledge before presenting your analysis.	CI/CD in CPS, embedded testing, mutation testing, digital twins, certification and graceful degradation.	Analytical, theme-based and comparative rather than source-by-source summaries.	Every major claim should be attached to a Harvard citation.
Methodology	Explains how you found, selected and analysed the 8 papers.	Databases, search terms, inclusion/exclusion criteria and comparison matrix fields.	Objective and reproducible; avoid overstating this as a full systematic review.	Cite Garousi et al. (2018) as an example of systematic mapping logic if useful.
Results / Comparative Analysis	Shows what the papers contain and how their structures/methods differ.	Section comparison, research approaches, data collection, sampling, analysis methods and findings.	Comparative and evidence-driven; use grouped patterns.	Use clusters of citations, e.g. (Zampetti et al., 2022; Dakkak et al., 2023).
Discussion	Evaluates what the findings mean for a risk-aware CI/CD pipeline.	Agreement, contradictions, extensions, gaps, implications and pipeline recommendations.	Most analytical section; link back to research question and objectives.	Use citations while making your own evaluation visible.

Section	Purpose	Typical content	Style	Citation cue
Conclusion	Answers the research question and closes the argument.	Main answer, contribution, limitations and future research direction.	Clear, restrained and no new evidence.	Citations are optional; avoid introducing new sources here.

Literature Matrix: Short Version

ID	Source	Methodology	Key finding	Gap	Use in article
P1	Zampetti et al. (2022)	Interview-based empirical software engineering study using semi-structured interviews, open card sorting, member checking and a validating survey with 55 professional developers.	CPS CI/CD requires a balance of continuous and periodic builds, careful use of simulators and Hardware-in-the-Loop, attention to deployment difficulty, and teams that combine hardware and software expertise.	Limited direct design of risk-aware rollback mechanisms; mostly organisational and practice-level evidence.	Use as the main anchor for why safety-critical embedded CI/CD must combine automation with periodic assurance gates.
P2	Dakkak et al. (2023)	Industrial case study at Ericsson AB focused on 3G Radio Access Network embedded software.	Deployment and field validation become continuous; orchestration between constituent systems, monitoring, documentation and	Not a safety-critical certification study; rollback is discussed less directly than monitoring and validation.	Use to explain staged deployment, field validation and monitoring as risk controls in complex embedded environments.

ID	Source	Methodology	Key finding	Gap	Use in article
			management support become essential.		
P3	Du et al. (2022)	Design science / engineering prototype evaluated on a real heterogeneous automotive system.	Multi-stage CI testing can reduce late defect discovery and support hardware-software co-design before final hardware is fully available.	Limited explicit treatment of rollback or certification evidence; focused on testing platform feasibility.	Use as concrete evidence for automated testing techniques suitable for embedded CI/CD pipelines.
P4	Baker and Habli (2013)	Empirical evaluation of mutation testing on safety-critical C and Ada airborne software.	Mutation testing can reveal weaknesses missed by traditional coverage and manual review, and can point to issues in requirements and coding.	Older study and not CI/CD-specific; needs linking to automated pipeline gates.	Use to argue that safety-critical CI pipelines should include stronger test adequacy checks, not just pass/fail unit tests.
P5	Garousi et al. (2018)	Systematic literature review / systematic literature mapping.	Embedded testing is broad and fragmented; practitioners need structured guidance to avoid reinventing testing approaches.	Survey is broad; it does not prescribe a specific risk-aware CI/CD architecture.	Use as the broad evidence base for selecting automated testing layers in the pipeline.
P6	Barbie and Hasselbring	Conceptual formalisation	Digital twin prototypes allow	Digital twin fidelity, safety	Use for simulation and

ID	Source	Methodology	Key finding	Gap	Use in article
	(2024)	using Object-Z and UML, with application in real-world field studies and a lab replication study.	virtual-context automated integration tests in CI/CD without constant access to the physical object.	certification acceptance and rollback integration remain open questions.	digital-twin testing as a scalable complement to HIL.
P7	Baron and Louis (2023)	Framework/tooling proposal illustrated through an industrial case study.	Certification requirements should be integrated continuously into the development lifecycle instead of treated as late audit work.	Avionics-specific; needs careful transfer to broader embedded domains like automotive or medical systems.	Use to support the argument that the pipeline must generate traceable assurance artefacts, not merely deploy software faster.
P8	Becker, Voss and Schätz (2018)	Formal modelling and constraint-based analysis using an SMT solver, applied to a constructed automotive example.	Safety-critical embedded systems need failure-handling actions such as fail-silent, fail-safe, fail-operational and degraded modes to avoid harmful system-level failure.	Formal model is not directly integrated with a CI/CD pipeline.	Use as the main theoretical basis for rollback, fail-safe and graceful degradation mechanisms.

Risk-Aware CI/CD Pipeline Model

Stage	Gate	Risk-control function
Stage 1	Commit and build gate	static checks, unit tests and

Stage	Gate	Risk-control function
		traceability checks.
Stage 2	Simulation / digital twin gate	fast repeatable integration tests using simulators or digital twin prototypes.
Stage 3	Hardware-in-the-Loop gate	selected tests on real or FPGA-supported hardware for timing and physical interaction risks.
Stage 4	Safety assurance gate	mutation/coverage evidence, requirements traceability and certification artefact checks.
Stage 5	Staged deployment gate	limited rollout, monitoring and field validation before wider release.
Stage 6	Rollback / fail-safe gate	version rollback, feature degradation, fail-silent/fail-safe/fail-operational mode selection.

Possible synthesis sentence: A risk-aware pipeline should treat rollback as one recovery option inside a broader assurance strategy, because safety-critical embedded systems may require degraded, fail-safe or fail-operational behaviour rather than a simple version reversal (Becker, Voss and Schätz, 2018).

Harvard In-Text Citation Guide

- Narrative citation: Zampetti et al. (2022) found that CPS CI/CD requires balancing simulators and Hardware-in-the-Loop testing.
- Parenthetical citation: CPS pipelines require both software and hardware expertise (Zampetti et al., 2022).
- Two-author citation: Baker and Habli (2013) showed that mutation testing can reveal weaknesses missed by coverage criteria.

- Three-author citation in this school pack: use Becker, Voss and Schätz (2018) for the first and later mentions unless your lecturer requires 'et al.' after first mention.
- Multiple sources: Automated embedded testing should combine broad testing knowledge, virtual environments and physical validation (Garousi et al., 2018; Du et al., 2022; Barbie and Hasselbring, 2024).
- Avoid unsupported claims. If a sentence contains a factual claim from the literature, cite it.

Drafting Scaffold

Abstract

- [One sentence on the deployment-risk problem.]
- [One sentence on your literature-search method: eight peer-reviewed studies were analysed.]
- [One sentence on the main pattern: layered testing + staged deployment + recovery design.]
- [One sentence on your recommendation.]

Introduction

- Open with why faster deployment is attractive but risky in safety-critical embedded systems.
- Contrast ordinary CI/CD with CPS constraints using Zampetti et al. (2022).
- Introduce system-of-systems validation pressure using Dakkak et al. (2023).
- End with your research question and objectives.

Methodology

- State databases: Google Scholar, IEEE Xplore, ACM Digital Library, ScienceDirect and institutional repositories.
- State search terms: CI/CD, continuous deployment, cyber-physical systems, embedded software testing, safety-critical software, mutation testing, Hardware-in-the-Loop, digital twin prototype, rollback, fail-safe and graceful degradation.

- State inclusion criteria: peer-reviewed, directly relevant to CI/CD/testing/safety/recovery, preferably 2018-2024 except foundational safety-critical testing evidence.
- State analysis method: each paper was compared by section structure, method, approach, framework, data collection, sampling, analysis, findings, contradictions and gaps.

Literature Review

- Theme 1: CI/CD in CPS and systems-of-systems.
- Theme 2: automated embedded testing, mutation testing, HIL, simulation and digital twins.
- Theme 3: continuous certification and safety assurance evidence.
- Theme 4: rollback, fail-safe behaviour and graceful degradation.

Results / Comparative Analysis

- Compare article structures: empirical studies have explicit method/results sections; formal/design papers emphasise models and evaluation; reviews emphasise search/selection methods.
- Compare methodologies: interviews/surveys, case studies, engineering prototypes, systematic mapping, formal modelling and conceptual modelling.
- Group findings into confirming, extending and contradicting patterns.

Discussion

- Answer the research question directly: automated testing reduces deployment risk by catching defects earlier and producing evidence, while rollback/fail-safe mechanisms reduce harm after faults escape.
- Evaluate gaps: limited integrated research connecting CI/CD, certification artefacts, HIL/digital twins and rollback in one pipeline.
- Propose the six-stage risk-aware CI/CD model from this pack.

Conclusion and Venue

- Close with a concise answer and recommendations.

- Name ICST because it covers software testing, verification, validation, empirical studies and embedded software.

Rubric Checklist

- 5-10 peer-reviewed sources included and all appear in the reference list.
- Common research paper sections are identified with purpose, content and writing style.
- Different article structures are compared instead of merely listed.
- Research methodologies, approaches, frameworks, data collection, sampling and analysis methods are identified.
- Findings are grouped into confirmations, extensions, contradictions, agreements and gaps.
- Recommendations connect directly to the chosen topic and research question.
- Harvard in-text citations match the final reference list.
- Final document uses Arial 12 and 1.5 spacing.

Harvard Reference List

Zampetti, F., Tamburri, D.A., Panichella, S., Panichella, A., Di Penta, M. and Canfora, G. (2022) 'Continuous integration and delivery practices for cyber-physical systems: An interview-based study', *ACM Transactions on Software Engineering and Methodology*, 32(3), Article 73. Available at: <https://doi.org/10.1145/3571854>.

Dakkak, A., Bosch, J., Holmström Olsson, H. and Mattos, D.I. (2023) 'Continuous deployment in software-intensive system-of-systems', *Information and Software Technology*, 159, Article 107200. Available at: <https://doi.org/10.1016/j.infsof.2023.107200>.

Du, B., Azimi, S., Moramarco, A., Sabena, D., Parisi, F. and Sterpone, L. (2022) 'An automated continuous integration multitest platform for automotive systems', *IEEE Systems Journal*, 16(2), pp. 2495-2506. Available at: <https://doi.org/10.1109/JSYST.2021.3069548>.

- Baker, R. and Habli, I. (2013) 'An empirical evaluation of mutation testing for improving the test quality of safety-critical software', IEEE Transactions on Software Engineering, 39(6), pp. 787-805. Available at: <https://doi.org/10.1109/TSE.2012.56>.
- Garousi, V., Felderer, M., Karapıçak, Ç.M. and Yılmaz, U. (2018) 'Testing embedded software: A survey of the literature', Information and Software Technology, 104, pp. 14-45. Available at: <https://doi.org/10.1016/j.infsof.2018.06.016>.
- Barbie, A. and Hasselbring, W. (2024) 'From digital twins to digital twin prototypes: Concepts, formalization, and applications', IEEE Access, 12, pp. 75337-75365. Available at: <https://doi.org/10.1109/ACCESS.2024.3406510>.
- Baron, C. and Louis, V. (2023) 'Framework and tooling proposals for Agile certification of safety-critical embedded software in avionic systems', Computers in Industry, 148, Article 103887. Available at: <https://doi.org/10.1016/j.compind.2023.103887>.
- Becker, K., Voss, S. and Schätz, B. (2018) 'Formal analysis of feature degradation in fault-tolerant automotive systems', Science of Computer Programming, 154, pp. 89-133. Available at: <https://doi.org/10.1016/j.scico.2017.10.007>.

Source Links

- P1 Zampetti et al. (2022): <https://research.tudelft.nl/en/publications/continuous-integration-and-delivery-practices-for-cyber-physical->
- P2 Dakkak et al. (2023): <https://research.tue.nl/en/publications/continuous-deployment-in-software-intensive-system-of-systems/>
- P3 Du et al. (2022): <https://iris.polito.it/handle/11583/2898672>
- P4 Baker and Habli (2013): <https://pure.york.ac.uk/portal/en/publications/an-empirical-evaluation-of-mutation-testing-for-improving-the-tes/>
- P5 Garousi et al. (2018): <https://research.wur.nl/en/publications/testing-embedded-software-a-survey-of-the-literature>
- P6 Barbie and Hasselbring (2024): https://www.cau-se.de/news/2024-06new_journal_paper_on_digital_twin_formalization/

P7 Baron and Louis (2023):

<https://www.sciencedirect.com/science/article/abs/pii/S0166361523000374>

P8 Becker, Voss and Schätz (2018):

<https://www.sciencedirect.com/science/article/pii/S0167642317302198>